

Tools Of The Trade

Due to incubation of robotics at the school levels, many elite schools from different corners of the country have participated in the World Robot Olympiad at the international level. At the elementary level, interest and proficiency in physics and mathematics prove to be beneficial in this domain.

“Organisations looking to recruit freshers prefer graduates like B.E. or B. Tech.,” says Saha. For development and research jobs, Masters and Ph.D. level students from the engineering domain as well as physics and mathematics are employed. Masters degree holders like M.E. or M. Tech also have a chance to apply and get through internship/training at the national labs or national level agencies which come out with such openings from time to time.

The chief educational streams to go through and the subjects to master are:

- Mechanical Engineering—kinematics, dynamics, design, simulation
- Electrical Engineering—electronics, controls
- Computer Engineering—control hardware, vision, embedded systems, artificial intelligence
- Information Technology—Programming for simulation and control

Top technical institutes provide different courses in robotics engineering or specialisation in subjects directly dealing with robotics. Scoring well in your engineering stream of interest by mastering the subjects mentioned will bring you a few steps closer to an exciting job. Taking part in technical festivals can also serve as an exclusive platform for display of skills and learning about new techniques applied in robotics.

Application And Training

The recruitment process involves the conventional procedure where more weightage is given to projects pursued by the applicants. Making robots from ready-made or semi-made kits is passé—students are expected to address more complex issues.



“Since a lot of automation activities are taking place in India to reduce defects and increase tolerances, there is a lot of scope for a career in robotics.”

Dr. Subir Kumar Saha
Professor
Dept. of Mechanical Engineering
IIT Delhi

Soft skills are also a must since candidates need to perform on an individual basis and also in a team. In addition to soft skills, freshers are required to have good cognitive skills, excellent analytical and problem solving abilities and reasoning abilities.

Various institutes offer very basic training along with the education while the students broaden their horizons by working on projects and taking part in tech-fests. Also quite a number of national labs/agencies and many private organisations like ABB Ltd., KUKA, FANUC, The Robotics Institute India, DiFACTO, etc. offer training services, courses and also workshops for professionals as well as enthusiasts.

Levels of hierarchy

The roles and responsibilities are well-defined in any lab or organisation working on robotics. Here the individuals do not have specific time schedule of working and most of the time end up working late.

- Service Engineers: Servicing and maintenance of existing robots
- Development Engineers: Developing new designs/methods etc. in specific areas for robotics in terms of hardware and software.
- System Engineers: Overall system integration, working closely with development teams
- R&D Engineers: Development of new technologies, application development.

Hierarchical structure for these different avenues remains the same. From trainee engineer or assistant engineer you can rise to become the director or vice president for that particular function or project. Freshers start as robot technicians and within two years advance to the next level in the relevant field of designing, development or controlling the robot. There are also robot controllers/operators who instruct robots.

Since majority of the work is project-based, the structure of an organisation dealing in robotics is similar to that of other engineering occupations.

Artificial Intelligence

“Robotics as a career has definitely a bright future in India. Development of robots for various critical purposes like GPS, navigation and also the human-like behaviour such as hearing, speech, and touch is also taking place at specialised scale,” says Mahalingam, “Even high schools are getting active with projects and expos for young and enthusiastic students.”

Forget all those sci-fi movies. Robotics has arrived. ■

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Payday

Designation	Salary Per Annum (INR)	Experience (Years)
Higher Technical & Administrative Positions	12 lakh and above	10 to 15
Project Manager / Engineering	8 to 12 lakh	5 to 10 or more
Senior Technician / Software Developer / Control Engineer	6.5 to 8 lakh	2 to 5 or more
Assts. Engineers / Technicians / Research Assistants	4 to 6 lakh	0 to 2